Syllabus - Math 15200, Section 25

Instructor: Miles Kretschmer

Fall Quarter 2026

This syllabus may be updated during the quarter, the most recent version will be on the Canvas page and on my website.

1 Information for Reference

- Course Website: Course information and assignments will be on the Canvas page, as well as on `https://mjkretschmer.github.io/152`.
- Instructor Information:
  - Email: `mjkretschmer@uchicago.edu`
    This is the best way to reach me. I am happy to respond to questions about the course over email.
  - Office: Eckhart 7, in the basement of Eckhart Hall
- Lectures: Monday, Wednesday, and Friday, 9:30 AM - 10:20 AM, in Eckhart 312
- Office Hours: Monday 2:30 - 3:30, Thursday and Friday 3:00 - 4:00. By default, these will be held in my office, but may move to E28A.
- Textbook: *Calculus: One and Several Variables*, 10th edition, Salas-Hille-Etgen. A pdf can be found on canvas and on my website.
- Homework: Each assignment will have two parts: a set of computational problems on the platform Edfinity, and a set of longer form problems to be turned in on Gradescope. Both parts will be accessed through Canvas, and will be assigned each Friday and due at 10:00 PM the next Friday. There will be 8 assignments in total.

  For policies regarding homework, see section 4.
- Exams: There will be two midterms and a final. The midterms are planned to take place at class time on Monday of week 4 and Wednesday of week 7, but may be pushed back. The final will take place on Thursday of finals week, from 8:00AM to 10:00 AM.

  For policies regarding exams, see sections 5 and 6.1.
- Grading: Homework will comprise 10% of the grade, each midterm will be 25%, and the final 40%.

  Letter grades will in all likelihood be determined by the following scale:

Letter	A	A-	B+	B	B-	C+	C	C-	D	F
Required Score	93	90	87	83	80	77	73	70	60	< 60

  It is possible that I will use a more lenient scale if exams are more difficult than anticipated.
- Prerequisites: This course will assume familiarity with the derivative, in particular its definition as a limit, its interpretation as a rate of change, and the rules to compute it.

2 Course Description and Objectives

This is a second course in calculus. We will begin by discussing limits, including the formal ε, δ definition, and reviewing differential calculus. From there, we will move on to the foundations of integration, the fundamental theorem of calculus, and basic integration techniques. On this basis, we will study applications of integration in solid geometry and physics. After that, we will see how transcendental functions such as the natural exponential and logarithm and inverse trigonometric functions arise from integrals of algebraic functions, and study more advanced techniques of integration including integration by parts and partial fraction decomposition. We will conclude by discussing linear differential equations. By the end of the quarter, you will be able to:

1. Construct a valid and precise mathematical argument, in particular using the formal definition of a limit.
2. Use the fundamental theorem of calculus to relate integration and differentiation, and evaluate integrals using a variety of techniques.
3. Express average values, areas, volumes, and accumulated changes in terms of integrals.
4. Use transcendental functions such as the logarithm, exponential, and inverse trigonometric functions to express certain integrals and solutions to equations.
5. Find solutions to certain linear differential equations using special transcendental functions and integration.

3 Planned Schedule

This is subject to change.

Week 1 9/28 - 10/2	Limits and Continuity (Chapter 2), Review of Differential Calculus (Chapters 3 and 4)	
Week 2 10/5 - 10/9	Definite Integrals (5.1-5.3), The Fundamental Theorem of Calculus and Indefinite Integrals (5.4-5.6)	10/7 Assignment 1 Due
Week 3 10/12 - 10/16	Properties of the Integral (5.8, 5.9), Integration by u-substitution (5.7)	10/14 Assignment 2 Due
Week 4 10/19 - 10/23	Areas of Plane Regions (5.5, 6.1), Volumes of Solids (6.2, 6.3)	10/19 Midterm 1 10/21 Assignment 3 Due
Week 5 10/26 - 10/30	More on Volume (6.2, 6.3), Average Values (5.9), Inverses of Functions (7.1)	10/28 Assignment 4 Due
Week 6 11/2 - 11/6	The Natural Logarithm (7.2, 7.3), The Exponential Function (7.4)	11/6 Assignment 5 Due
Week 7 11/9 - 11/13	Inverse Trigonometric Functions (7.7), Integration by Parts (8.2)	11/11 Midterm 2 11/13 Assignment 6 Due
Week 8 11/16 - 11/20	Trigonometric Identities and Substitution (8.3-8.4), Integration of Rational Functions (8.5)	11/20 Assignment 7 Due
Thanksgiving break		
Week 9 11/30 - 12/4	Linear Differential Equations (Chapter 9)	12/4 Assignment 8 Due
Finals week		Final Thursday 12/10 8:00 AM - 10:00 AM

4 Homework Policies

The purpose of the problem sets is to give you practice with the things introduced in lecture, and working through them will be a necessary part of learning the material of the course. Because of this, I ask you to not use AI when working on them. You are welcome to discuss problems with other students, but you should write up your own solutions in your own words, and make sure you can carry out computations on your own.

4.1 Late Policy

You can submit one assignment up to two days late, which comes out to 10:00 PM on Friday. This applies to both parts of the assignment. If you will submit an assignment late, email me to let me know of this by the Wednesday that it is due. Outside of this, late work will not be accepted.

5 Exam Policies

Exams will consist mostly of computational problems similar to problems assigned for homework. In week 9, I will release a set of practice problems for the final. The exams will be closed book, with no electronic devices permitted. Solutions should come with justification for full credit. Partial credit will be given for partial solutions. Please let me know as soon as possible if some exceptional circumstance prevents you from taking the exam. In normal circumstances, there will be no make up exams.

6 University Policies

Add/drop: Add/drop for all courses ends Friday Week 3, on 10/16. This is the last day you can drop a class without it being present on your academic transcript.
Withdraw (W): After the add/drop deadline, you can still withdraw from the course up to Monday, 11/30 in Week 9 at 5pm, but it will result in a W (withdraw) grade on your transcript.
Incomplete (I): If emergency circumstances prevent you from taking the final exam, you may be eligible to request an "I" (incomplete). A request form must be signed by your academic adviser and your instructor.
Pass/fail: Pass/fail grades are not permitted for this course.

6.1 Final Exam

It is the policy of the Department of Mathematics that the following rules apply to final exams in all undergraduate mathematics courses:
The final exam must occur at the time and place designated on the College Final Exam Schedule. In particular, no final examinations may be given during the Ninth Week of the quarter. Instructors are not permitted to excuse students from the scheduled time of the final exam except in the cases of an Incomplete.

6.2 Disability Accommodations

Students with disabilities who have been approved for the use of academic accommodations by Student Disability Services (SDS) and need a reasonable accommodation(s) to participate fully in this course should follow the procedures established by SDS for using accommodations. Timely notifications are required in order to ensure that your accommodations can be implemented. Please meet with me to discuss your access needs in this class after you have completed the SDS procedures for requesting accommodations. SDS can be reached at:
Phone: `(773) 702-6000`
Email: `disabilities@uchicago.edu`